Natural Language processing

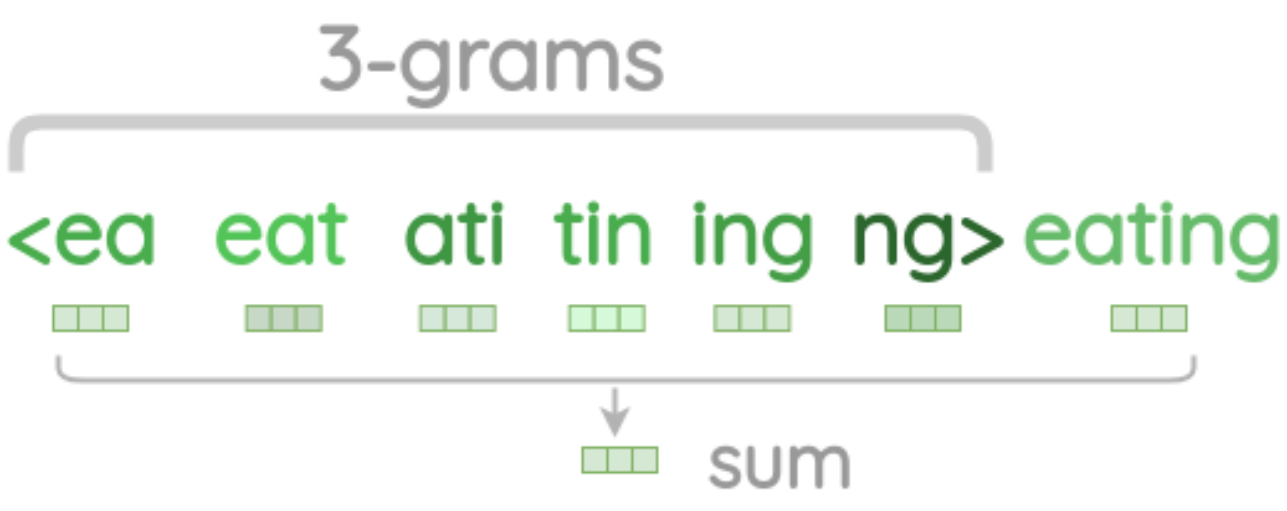
Section 4

‘Word Embedding using FastText

FastText

- **FastText** embeddings are a type of word embedding developed by Facebook's AI Research (FAIR) lab. They are based on the idea of subword embeddings, which means that instead of representing words as single entities, **FastText** breaks them down into smaller components called **character n-grams**.
- By doing so, **FastText** can capture the semantic meaning of morphologically related words, even for out-of-vocabulary words or rare words, making it particularly useful for handling languages with **rich morphology**

FastText



Code Implementation of FastText Embeddings

```python
from gensim.models import FastText
from gensim.test.utils import common_texts

# Example corpus (replace with your own corpus)
corpus = common_texts

# Training FastText model
model = FastText(sentences=corpus, vector_size=100, window=5, min_count=1, workers=4, sg=1)

# Example usage: getting embeddings for a word
word_embedding = model.wv['computer']

# Most similar words to a given word
similar_words = model.wv.most_similar('computer')

print("Most similar words to 'computer':", similar_words)
```

```
Most similar words to 'computer': [('user', 0.15659411251544952), ('response', 0.12383826076984406), ('eps', 0.030704911798238754), ('system', 0.02557388320565223 7), ('interface', 0
.005858752410858869 6), ('survey', -0.0315697640180587 8), ('minors', -0.054556474089622 5), ('human', -0.0668589174747467), ('time', -0.06855931878089905), ('trees', -0.10636083036661
148)]
```

See you soon